Supplementary Tables

Label-efficient choice of proxy tracks for variant-effect prediction

All tables are generated by `paper/make_supplement.py` from the result tables under `results/a1` in the code repository. Spearman correlations are between predicted and measured effects of held-out variants; differences are paired within a unit (Kircher genomic region, or Siraj locus with its two cell types averaged). Siraj-derived tables are released under CC-BY-NC 4.0. Table S6 has two parts (S6a, S6b).

Table S1: Siraj test (125 loci): effects relative to the measured reference oligo (primary) and centred on all oligos of the element as published (original analysis). P1–P4 are the pre-registered hypotheses; all held under both definitions. Secondary rows give the mean difference over loci and its 95% bootstrap interval. S3b was added with the reference-relative labels.

	Quantity	Reference-relative	Published centring
P1 $k = 8$	mean D (lower bound)	+0.009 (+0.005)	+0.010 (+0.005)
P2 $k = 8$	harmed (upper bound)	3/125 (0.061)	3/125 (0.061)
P1 $k = 16$	mean D (lower bound)	+0.019 (+0.013)	+0.020 (+0.013)
P2 $k = 16$	harmed (upper bound)	1/125 (0.037)	1/125 (0.037)
P3 $k = 8$	v2 − all-track mean (P)	+0.047 (P=3e-12)	+0.048 (P=8e-13)
P4 $k = 16$	v2 − related mean (P)	+0.040 (P=3e-19)	+0.041 (P=1e-19)
S1 C1 k=8	v2 disc panel − v2 random panel	+0.016 [+0.012 to +0.021]	+0.018 [+0.013 to +0.022]
S1 C3 k=8	v2 disc panel − v2 random panel	+0.096 [+0.087 to +0.104]	+0.093 [+0.085 to +0.101]
S2 C1 k=8	v2 − meta_ridge on the same disc panel	+0.007 [+0.003 to +0.011]	+0.008 [+0.004 to +0.012]
S3 C1 k=8	matched_cal (random panel) − matched	−0.082 [−0.090 to −0.075]	−0.079 [−0.086 to −0.072]
S3b C1 k=8	matched_cal (disc panel) − matched	−0.010 [−0.015 to −0.006]	–
S5 C1 k=8	family prior − uniform prior	−0.002 [−0.004 to −0.000]	−0.002 [−0.003 to +0.000]
S7 HEPG2 C1 k=8	v2 − matched	+0.016 [+0.007 to +0.026]	+0.017 [+0.007 to +0.026]
S7 HEPG2 C3 k=8	v2 − all_mean	+0.050 [+0.029 to +0.074]	+0.052 [+0.031 to +0.076]
S7 K562 C1 k=8	v2 − matched	+0.003 [−0.001 to +0.007]	+0.003 [−0.002 to +0.007]
S7 K562 C3 k=8	v2 − all_mean	+0.043 [+0.029 to +0.058]	+0.044 [+0.031 to +0.059]

Table S4: Acquisition comparators (post hoc; frozen v2 inference, 50 repeats). Siraj views C1–C3; Kircher test regions with matched tracks kept (deploy) or without an anchor (no_anchor). Reading: 95% interval within ±0.02 = equivalent; above or below 0 = first or second better.

Data	View	k	Contrast	n	Mean	95% CI	Reading
Siraj	C1	8	disc − batch_kb	125	+0.002	−0.000 to +0.004	equivalent

Data	View	k	Contrast	n	Mean	95% CI	Reading
Siraj	C2	8	disc − batch_kb	125	+0.005	+0.003 to +0.007	equivalent
Siraj	C3	8	disc − batch_kb	125	+0.004	+0.002 to +0.006	equivalent
Siraj	C1	8	disc − disc_static	125	+0.007	+0.004 to +0.012	equivalent
Siraj	C2	8	disc − disc_static	125	+0.014	+0.008 to +0.020	first better
Siraj	C3	8	disc − disc_static	125	+0.023	+0.017 to +0.029	first better
Siraj	C1	8	eig − disc	125	+0.000	−0.001 to +0.002	equivalent
Siraj	C2	8	eig − disc	125	+0.001	−0.000 to +0.002	equivalent
Siraj	C3	8	eig − disc	125	+0.000	−0.001 to +0.002	equivalent
Siraj	C1	8	batch_kb − disc_static	125	+0.005	+0.002 to +0.010	equivalent
Siraj	C2	8	batch_kb − disc_static	125	+0.009	+0.004 to +0.014	equivalent
Siraj	C3	8	batch_kb − disc_static	125	+0.019	+0.014 to +0.025	first better
Siraj	C1	8	disc − high_effect	125	+0.010	+0.005 to +0.016	equivalent
Siraj	C2	8	disc − high_effect	125	+0.017	+0.011 to +0.024	first better
Siraj	C3	8	disc − high_effect	125	+0.028	+0.020 to +0.035	first better
Siraj	C1	8	disc − disagree	125	+0.013	+0.008 to +0.018	equivalent
Siraj	C2	8	disc − disagree	125	+0.017	+0.011 to +0.023	first better
Siraj	C3	8	disc − disagree	125	+0.032	+0.024 to +0.041	first better
Siraj	C1	16	disc − batch_kb	125	+0.004	+0.002 to +0.007	equivalent
Siraj	C2	16	disc − batch_kb	125	+0.008	+0.006 to +0.010	equivalent
Siraj	C3	16	disc − batch_kb	125	+0.006	+0.004 to +0.009	equivalent
Siraj	C1	16	disc − disc_static	125	+0.009	+0.005 to +0.013	equivalent
Siraj	C2	16	disc − disc_static	125	+0.016	+0.013 to +0.021	first better
Siraj	C3	16	disc − disc_static	125	+0.016	+0.012 to +0.021	first better
Siraj	C1	16	eig − disc	125	+0.001	+0.000 to +0.002	equivalent
Siraj	C2	16	eig − disc	125	+0.001	+0.000 to +0.002	equivalent
Siraj	C3	16	eig − disc	125	+0.000	−0.001 to +0.001	equivalent
Siraj	C1	16	batch_kb − disc_static	125	+0.005	+0.002 to +0.009	equivalent
Siraj	C2	16	batch_kb − disc_static	125	+0.008	+0.005 to +0.012	equivalent
Siraj	C3	16	batch_kb − disc_static	125	+0.010	+0.006 to +0.014	equivalent
Siraj	C1	16	disc − high_effect	125	+0.012	+0.007 to +0.018	equivalent
Siraj	C2	16	disc − high_effect	125	+0.023	+0.018 to +0.029	first better
Siraj	C3	16	disc − high_effect	125	+0.027	+0.021 to +0.033	first better
Siraj	C1	16	disc − disagree	125	+0.011	+0.006 to +0.016	equivalent
Siraj	C2	16	disc − disagree	125	+0.015	+0.011 to +0.020	first better
Siraj	C3	16	disc − disagree	125	+0.020	+0.016 to +0.025	first better
Kircher	deploy	8	disc − batch_kb	11	+0.004	−0.001 to +0.009	equivalent
Kircher	no_anchor	8	disc − batch_kb	11	+0.003	−0.004 to +0.010	equivalent
Kircher	deploy	8	disc − disc_static	11	+0.006	+0.000 to +0.011	equivalent
Kircher	no_anchor	8	disc − disc_static	11	+0.015	−0.001 to +0.035	undetermined
Kircher	deploy	8	eig − disc	11	+0.003	+0.001 to +0.005	equivalent
Kircher	no_anchor	8	eig − disc	11	+0.002	−0.001 to +0.005	equivalent
Kircher	deploy	8	batch_kb − disc_static	11	+0.002	−0.003 to +0.006	equivalent
Kircher	no_anchor	8	batch_kb − disc_static	11	+0.012	+0.001 to +0.029	first better
Kircher	deploy	8	disc − high_effect	11	+0.004	−0.007 to +0.017	equivalent

Data	View	k	Contrast	n	Mean	95% CI	Reading
Kircher	no_anchor	8	disc − high_effect	11	+0.029	+0.001 to +0.059	first better
Kircher	deploy	8	disc − disagree	11	+0.005	−0.009 to +0.023	undetermined
Kircher	no_anchor	8	disc − disagree	11	+0.019	−0.003 to +0.041	undetermined
Kircher	deploy	16	disc − batch_kb	11	+0.005	+0.000 to +0.012	equivalent
Kircher	no_anchor	16	disc − batch_kb	11	+0.008	+0.002 to +0.013	equivalent
Kircher	deploy	16	disc − disc_static	11	+0.020	+0.000 to +0.048	first better
Kircher	no_anchor	16	disc − disc_static	11	+0.013	−0.002 to +0.028	undetermined
Kircher	deploy	16	eig − disc	11	+0.004	+0.001 to +0.007	equivalent
Kircher	no_anchor	16	eig − disc	11	+0.002	−0.001 to +0.006	equivalent
Kircher	deploy	16	batch_kb − disc_static	11	+0.015	−0.003 to +0.043	undetermined
Kircher	no_anchor	16	batch_kb − disc_static	11	+0.005	−0.011 to +0.018	equivalent
Kircher	deploy	16	disc − high_effect	11	+0.022	−0.004 to +0.055	undetermined
Kircher	no_anchor	16	disc − high_effect	11	+0.001	−0.021 to +0.023	undetermined
Kircher	deploy	16	disc − disagree	11	+0.020	+0.002 to +0.041	first better
Kircher	no_anchor	16	disc − disagree	11	+0.008	−0.004 to +0.019	equivalent

Table S5: CODA- and MODEL SELECTOR-style baselines (post hoc; 50 repeats). Both use the calibrated likelihood, no metadata and the information-gain panel; CODA uses a consensus prior ($\kappa = 10$, fixed on development elements), MODEL SELECTOR a uniform prior ($\kappa = 0$). *Single track*: the most probable track, the methods' own output; *average*: their posterior average. Positive: units where the first is better.

Data	View	k	Contrast	n	Mean	95% CI	+	Reading
Siraj	C1	8	v2 − coda (single track)	125	+0.075	+0.062 to +0.089	111	first better
Siraj	C1	8	v2 − model_selector (single track)	125	+0.074	+0.064 to +0.084	118	first better
Siraj	C1	8	coda: average − single track	125	+0.016	+0.012 to +0.021	99	first better
Siraj	C1	8	v2 − coda (average)	125	+0.059	+0.045 to +0.072	105	first better
Siraj	C1	16	v2 − coda (single track)	125	+0.045	+0.038 to +0.054	111	first better
Siraj	C1	16	v2 − model_selector (single track)	125	+0.042	+0.035 to +0.049	112	first better
Siraj	C1	16	coda: average − single track	125	+0.009	+0.005 to +0.012	92	equivalent
Siraj	C1	16	v2 − coda (average)	125	+0.037	+0.028 to +0.045	107	first better
Siraj	C2	8	v2 − coda (single track)	125	+0.029	+0.019 to +0.039	88	first better
Siraj	C2	8	v2 − model_selector (single track)	125	+0.036	+0.029 to +0.045	99	first better
Siraj	C2	8	coda: average − single track	125	+0.018	+0.014 to +0.022	103	first better
Siraj	C2	8	v2 − coda (average)	125	+0.011	+0.002 to +0.021	70	first better
Siraj	C2	16	v2 − coda (single track)	125	+0.017	+0.010 to +0.023	88	first better
Siraj	C2	16	v2 − model_selector (single track)	125	+0.018	+0.013 to +0.023	90	first better
Siraj	C2	16	coda: average − single track	125	+0.012	+0.009 to +0.015	98	equivalent
Siraj	C2	16	v2 − coda (average)	125	+0.005	−0.001 to +0.011	70	equivalent
Siraj	C3	8	v2 − coda (single track)	125	+0.034	+0.027 to +0.042	106	first better
Siraj	C3	8	v2 − model_selector (single track)	125	+0.044	+0.039 to +0.049	121	first better
Siraj	C3	8	coda: average − single track	125	+0.017	+0.013 to +0.021	105	first better
Siraj	C3	8	v2 − coda (average)	125	+0.018	+0.011 to +0.024	83	first better
Siraj	C3	16	v2 − coda (single track)	125	+0.028	+0.023 to +0.034	114	first better

Data	View	k	Contrast	n	Mean	95% CI	+	Reading
Siraj	C3	16	v2 − model_selector (single track)	125	+0.030	+0.026 to +0.034	115	first better
Siraj	C3	16	coda: average − single track	125	+0.012	+0.009 to +0.015	98	equivalent
Siraj	C3	16	v2 − coda (average)	125	+0.016	+0.012 to +0.021	93	first better
Kircher	C1	8	v2 − coda (single track)	11	+0.082	+0.006 to +0.149	8	first better
Kircher	C1	8	v2 − model_selector (single track)	11	+0.070	−0.003 to +0.138	8	undetermined
Kircher	C1	8	coda: average − single track	11	+0.026	+0.015 to +0.037	9	first better
Kircher	C1	8	v2 − coda (average)	11	+0.056	−0.019 to +0.120	8	undetermined
Kircher	C1	16	v2 − coda (single track)	11	+0.056	+0.013 to +0.099	9	first better
Kircher	C1	16	v2 − model_selector (single track)	11	+0.061	+0.013 to +0.108	7	first better
Kircher	C1	16	coda: average − single track	11	+0.017	+0.005 to +0.029	7	first better
Kircher	C1	16	v2 − coda (average)	11	+0.039	−0.007 to +0.081	8	undetermined
Kircher	C3	8	v2 − coda (single track)	11	+0.051	+0.018 to +0.087	8	first better
Kircher	C3	8	v2 − model_selector (single track)	11	+0.042	+0.005 to +0.075	8	first better
Kircher	C3	8	coda: average − single track	11	+0.025	+0.015 to +0.034	9	first better
Kircher	C3	8	v2 − coda (average)	11	+0.026	−0.005 to +0.059	7	undetermined
Kircher	C3	16	v2 − coda (single track)	11	+0.030	+0.015 to +0.045	9	first better
Kircher	C3	16	v2 − model_selector (single track)	11	+0.028	+0.002 to +0.051	8	first better
Kircher	C3	16	coda: average − single track	11	+0.019	+0.007 to +0.032	8	first better
Kircher	C3	16	v2 − coda (average)	11	+0.011	−0.008 to +0.028	6	undetermined

Table S6b: EIG convergence check, accuracy: differences in Spearman correlation of the posterior-average prediction (same pools and 50 repeats; 95% bootstrap interval over units).

Data	View	k	Contrast	n	Mean	95% CI	Reading
Siraj	C1	8	E2 − E0	125	−0.0002	−0.001 to +0.001	equivalent
Siraj	C2	8	E2 − E0	125	−0.0002	−0.001 to +0.001	equivalent
Siraj	C3	8	E2 − E0	125	+0.0004	−0.001 to +0.001	equivalent
Siraj	C1	8	E2 − disc	125	−0.0000	−0.001 to +0.001	equivalent
Siraj	C2	8	E2 − disc	125	+0.0006	−0.001 to +0.002	equivalent
Siraj	C3	8	E2 − disc	125	+0.0007	−0.001 to +0.002	equivalent
Siraj	C1	8	E0 − disc	125	+0.0002	−0.001 to +0.002	equivalent
Siraj	C2	8	E0 − disc	125	+0.0008	−0.000 to +0.002	equivalent
Siraj	C3	8	E0 − disc	125	+0.0003	−0.001 to +0.002	equivalent
Siraj	C1	8	E1 − E0	125	−0.0001	−0.001 to +0.001	equivalent
Siraj	C2	8	E1 − E0	125	−0.0002	−0.001 to +0.001	equivalent
Siraj	C3	8	E1 − E0	125	+0.0003	−0.001 to +0.001	equivalent
Siraj	C1	16	E2 − E0	125	−0.0005	−0.001 to +0.000	equivalent
Siraj	C2	16	E2 − E0	125	−0.0002	−0.001 to +0.001	equivalent
Siraj	C3	16	E2 − E0	125	−0.0001	−0.001 to +0.001	equivalent
Siraj	C1	16	E2 − disc	125	+0.0007	+0.000 to +0.001	equivalent
Siraj	C2	16	E2 − disc	125	+0.0009	+0.000 to +0.002	equivalent
Siraj	C3	16	E2 − disc	125	+0.0001	−0.000 to +0.001	equivalent
Siraj	C1	16	E0 − disc	125	+0.0012	+0.000 to +0.002	equivalent

Data	View	k	Contrast	n	Mean	95% CI	Reading
Siraj	C2	16	E0 − disc	125	+0.0011	+0.000 to +0.002	equivalent
Siraj	C3	16	E0 − disc	125	+0.0001	−0.001 to +0.001	equivalent
Siraj	C1	16	E1 − E0	125	−0.0005	−0.001 to +0.000	equivalent
Siraj	C2	16	E1 − E0	125	−0.0002	−0.001 to +0.000	equivalent
Siraj	C3	16	E1 − E0	125	+0.0000	−0.001 to +0.001	equivalent
Kircher	deploy	8	E2 − E0	11	+0.0007	−0.001 to +0.002	equivalent
Kircher	no_anchor	8	E2 − E0	11	−0.0001	−0.002 to +0.002	equivalent
Kircher	deploy	8	E2 − disc	11	+0.0038	+0.002 to +0.006	equivalent
Kircher	no_anchor	8	E2 − disc	11	+0.0022	−0.000 to +0.004	equivalent
Kircher	deploy	8	E0 − disc	11	+0.0031	+0.001 to +0.005	equivalent
Kircher	no_anchor	8	E0 − disc	11	+0.0023	−0.001 to +0.005	equivalent
Kircher	deploy	8	E1 − E0	11	+0.0012	−0.000 to +0.003	equivalent
Kircher	no_anchor	8	E1 − E0	11	−0.0003	−0.003 to +0.002	equivalent
Kircher	deploy	16	E2 − E0	11	−0.0010	−0.004 to +0.002	equivalent
Kircher	no_anchor	16	E2 − E0	11	−0.0001	−0.004 to +0.003	equivalent
Kircher	deploy	16	E2 − disc	11	+0.0029	+0.000 to +0.006	equivalent
Kircher	no_anchor	16	E2 − disc	11	+0.0022	+0.001 to +0.004	equivalent
Kircher	deploy	16	E0 − disc	11	+0.0038	+0.001 to +0.007	equivalent
Kircher	no_anchor	16	E0 − disc	11	+0.0023	−0.001 to +0.006	equivalent
Kircher	deploy	16	E1 − E0	11	+0.0000	−0.003 to +0.004	equivalent
Kircher	no_anchor	16	E1 − E0	11	−0.0013	−0.006 to +0.002	equivalent

Table S2: Directional accuracy (post hoc): fraction of significant held-out variants ($P < 10^{-5}$, |effect| ≥ 0.1) whose predicted sign is correct. Labelled methods use the posterior-average prediction; *no offset* drops the fitted offsets (diagnostic). Interval: 95% bootstrap over units; last column, units better/worse.

Set	Contrast	First	Second	n	Acc. 1	Acc. 2	Diff.	95% CI	+/−
Siraj	P1	disc@8	matched@0	125	0.874	0.867	+0.007	−0.010 to +0.024	63/60
Siraj	P1 (S4)	disc@16	matched@0	125	0.886	0.867	+0.019	+0.001 to +0.037	66/57
Siraj	P3	disc@8	all_mean@0	125	0.864	0.809	+0.055	+0.032 to +0.078	76/48
Siraj	P4	disc@16	related@0	125	0.873	0.834	+0.039	+0.020 to +0.057	80/45
Siraj	S1	disc@8	random@8	125	0.864	0.789	+0.074	+0.061 to +0.089	107/18
Siraj	S3	random+matched_cal@8	matched@0	125	0.817	0.867	−0.050	−0.068 to −0.032	30/95
Siraj	S3b	disc+matched_cal@8	matched@0	125	0.865	0.867	−0.002	−0.021 to +0.017	57/66
Kircher (v1)	H1	disc@8	random@8	11	0.675	0.694	−0.018	−0.105 to +0.042	7/4
Kircher (v1)	H2	disc@8	all_mean@0	11	0.675	0.774	−0.099	−0.267 to +0.046	4/7
Kircher (v2)	v2 C1 k=8	disc@8	matched@0	11	0.706	0.839	−0.133	−0.264 to −0.026	2/9
Kircher (v2)	v2 C1 k=16	disc@16	matched@0	11	0.730	0.839	−0.108	−0.230 to −0.010	3/8
Kircher (v2, no offset)	v2 C1 k=8	disc@8	matched@0	11	0.768	0.839	−0.071	−0.207 to +0.011	5/4
Kircher (v2, no offset)	v2 C1 k=16	disc@16	matched@0	11	0.790	0.839	−0.048	−0.158 to +0.027	6/3
Kircher (v2, no offset)	v2 C3 k=8	disc@8	all_mean@0	11	0.743	0.774	−0.031	−0.188 to +0.107	8/3
Kircher (v2, no offset)	v2 C3 k=8 vs random	disc@8	random@8	11	0.743	0.700	+0.042	−0.029 to +0.101	9/2
Kircher (v2)	v2 C3 k=8	disc@8	all_mean@0	11	0.694	0.774	−0.079	−0.232 to +0.055	4/7
Kircher (v2)	v2 C3 k=8 vs random	disc@8	random@8	11	0.694	0.693	+0.002	−0.066 to +0.052	8/3

Table S3: Siraj test repeated with, for each locus, the AlphaGenome checkpoint that held out its genomic fold (labels relative to the reference oligo). Columns give the pre-registered contrast with the distilled checkpoint (ALL_FOLDS) and with the held-out checkpoint, and the paired change with its 95% bootstrap interval over loci.

Hypothesis	ALL_FOLDS	Held-out fold	Change [95% CI]	Verdict
P1 $k = 8$	+0.009	+0.024	+0.014 [+0.008 to +0.022]	holds
P1 $k = 16$	+0.019	+0.037	+0.018 [+0.009 to +0.027]	holds
P3 $k = 8$	+0.047	+0.050	+0.003 [−0.006 to +0.013]	holds
P4 $k = 16$	+0.040	+0.053	+0.013 [+0.006 to +0.021]	holds
P2 $k = 8$	3/125 harmed	1/125 harmed	–	holds
P2 $k = 16$	1/125 harmed	1/125 harmed	–	holds

Table S6a: EIG convergence check (post hoc), coverage: posterior mass held by the hypotheses kept, median over units (minimum), at the first step and the lowest over the eight steps of a $k = 8$ panel. E0: the 100 most probable hypotheses, 96 quadrature points (as in the main text); E1: 300 hypotheses; E2 (not shown) keeps ≥99% of the mass at every step with 192 points.

Data	View	E0 first	E0 lowest	E1 first	E1 lowest
Siraj	C1	0.98 (0.96)	0.93 (0.79)	0.99 (0.98)	0.98 (0.91)
Siraj	C2	0.98 (0.96)	0.94 (0.76)	0.99 (0.98)	0.98 (0.91)
Siraj	C3	0.61 (0.23)	0.60 (0.23)	0.87 (0.66)	0.86 (0.66)
Kircher	deploy	0.97 (0.97)	0.95 (0.79)	0.99 (0.99)	0.98 (0.92)
Kircher	no_anchor	0.49 (0.32)	0.49 (0.32)	0.84 (0.76)	0.84 (0.76)